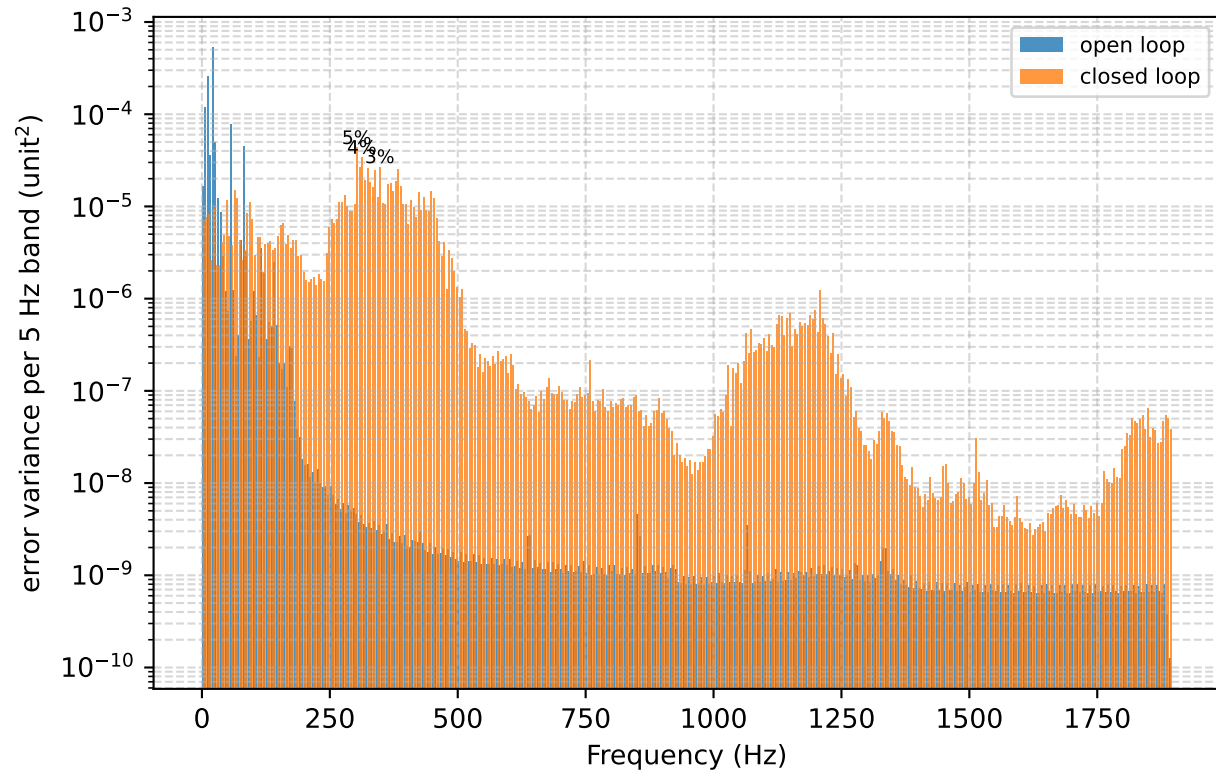
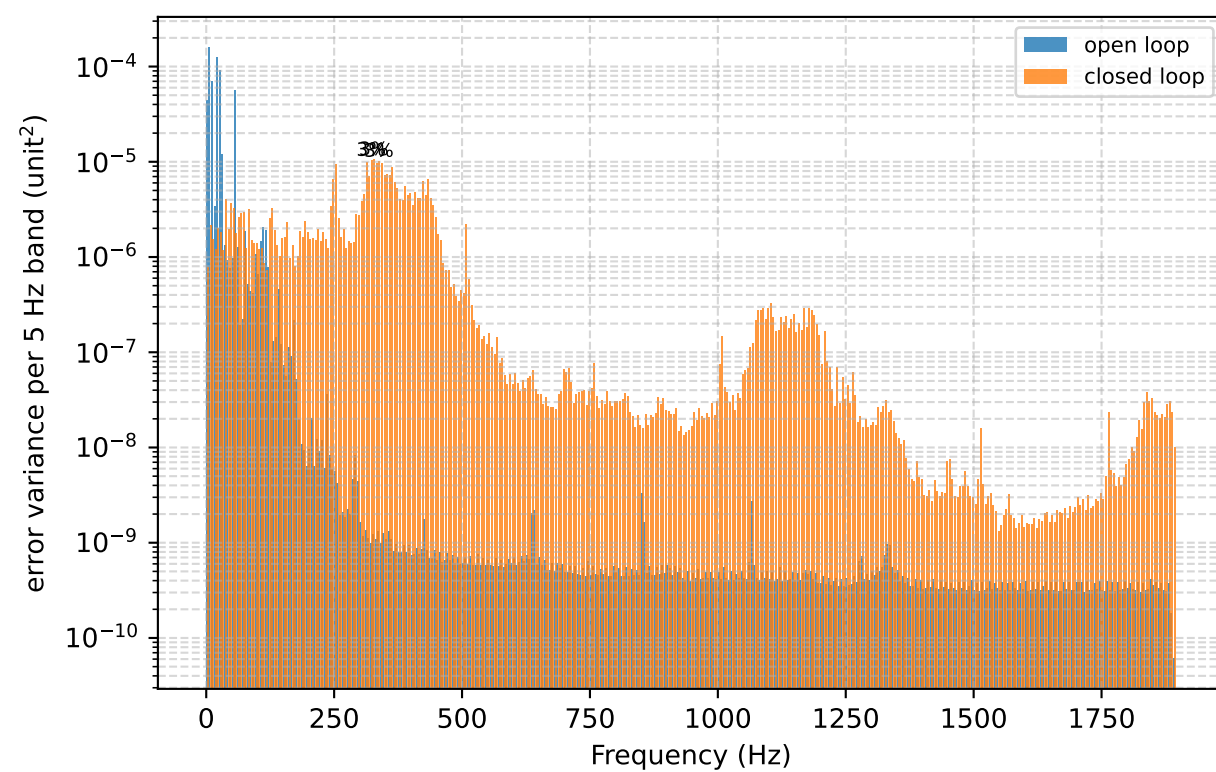
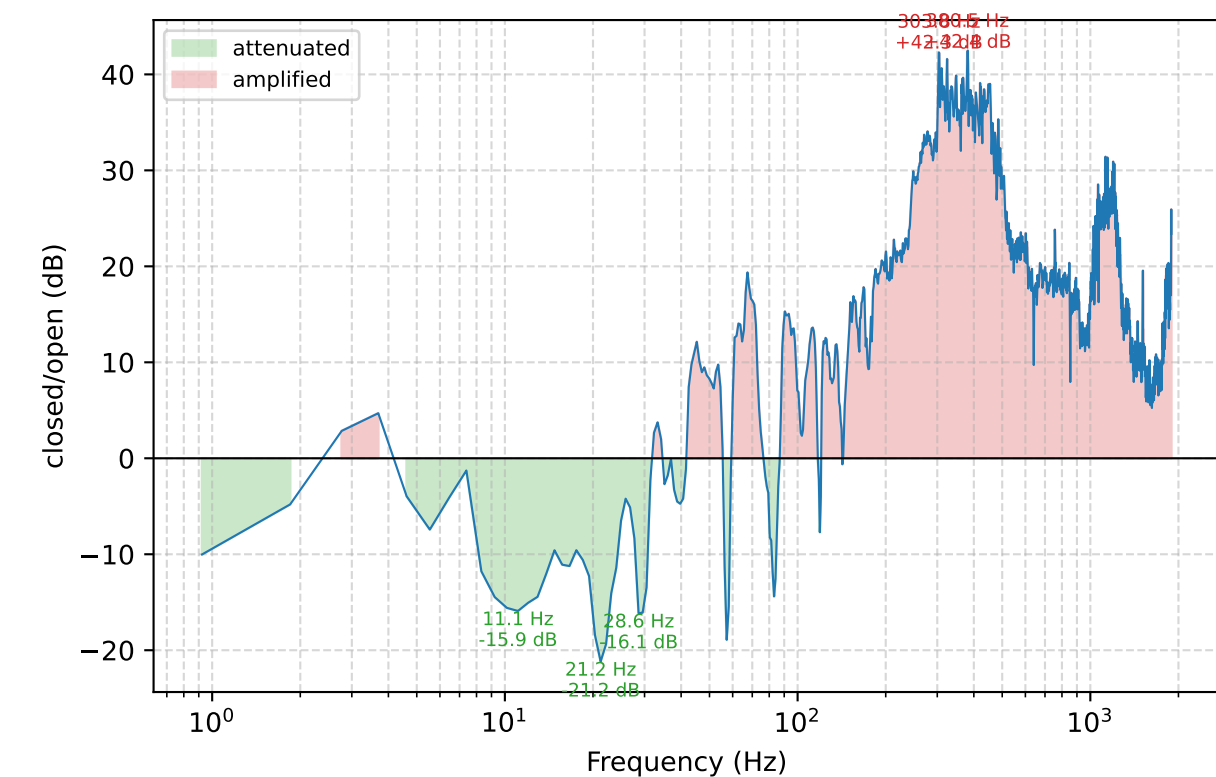
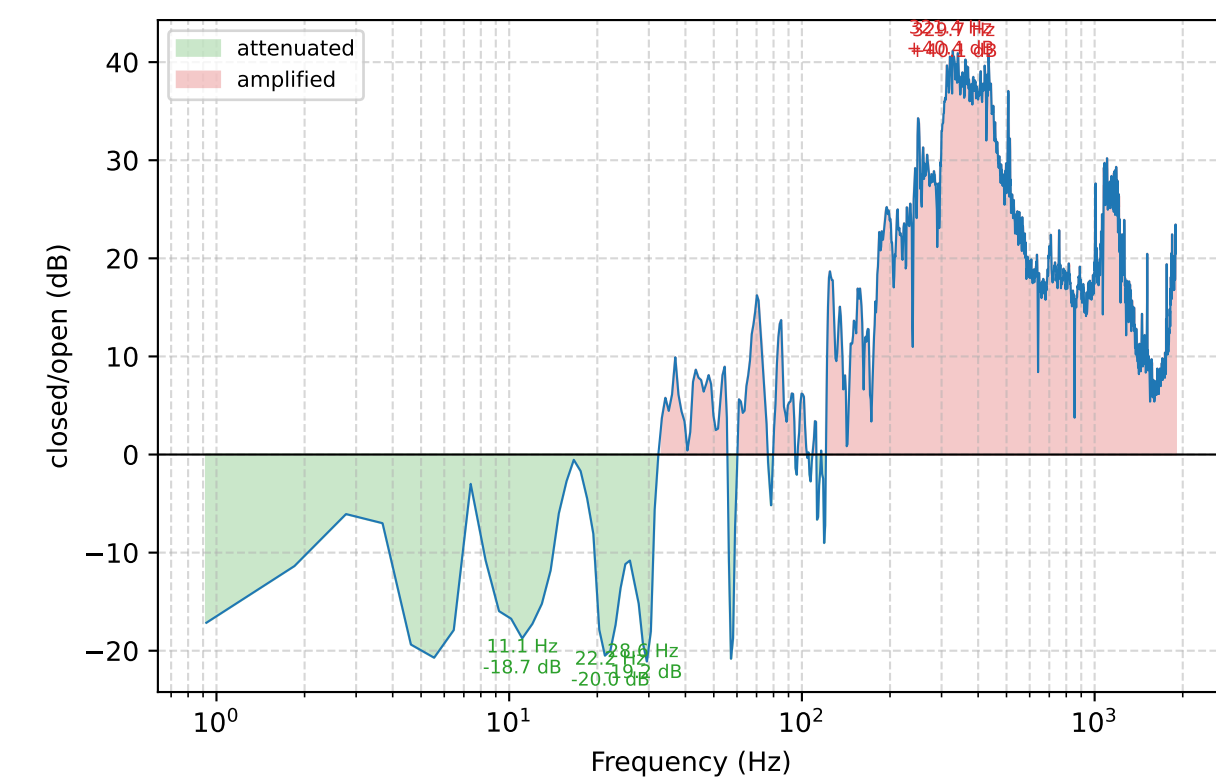
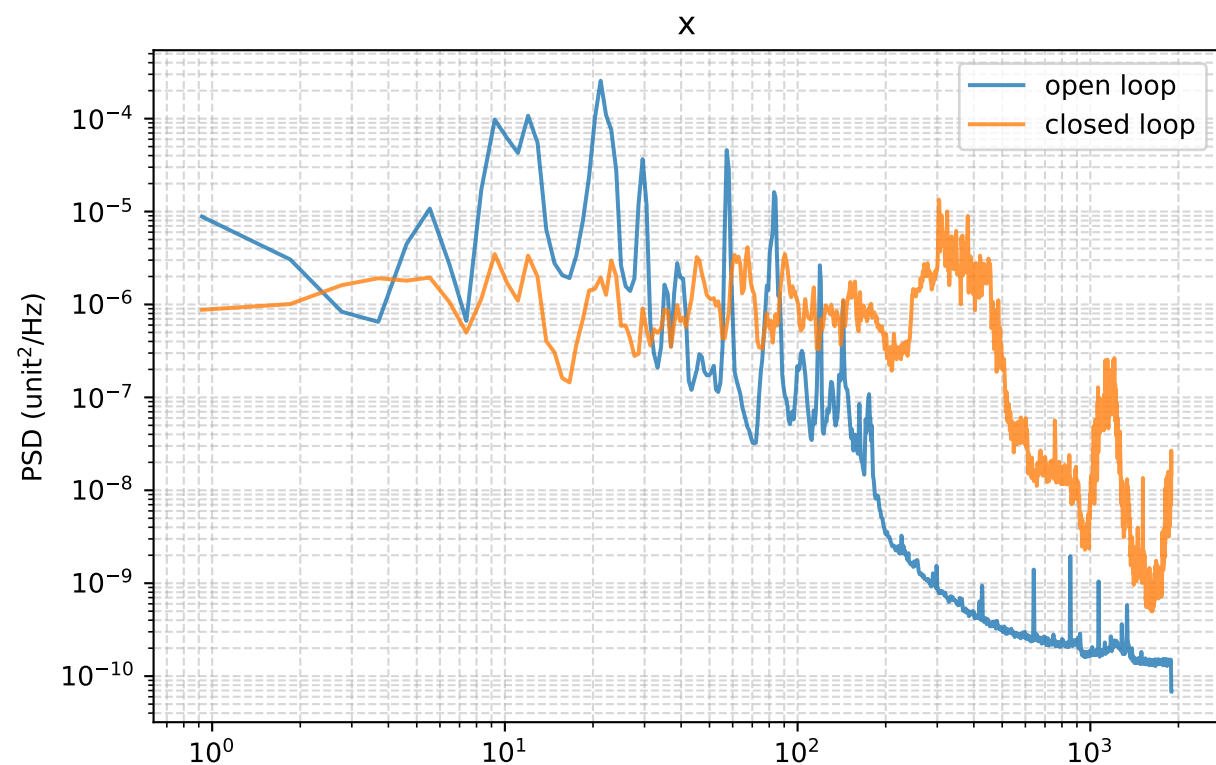
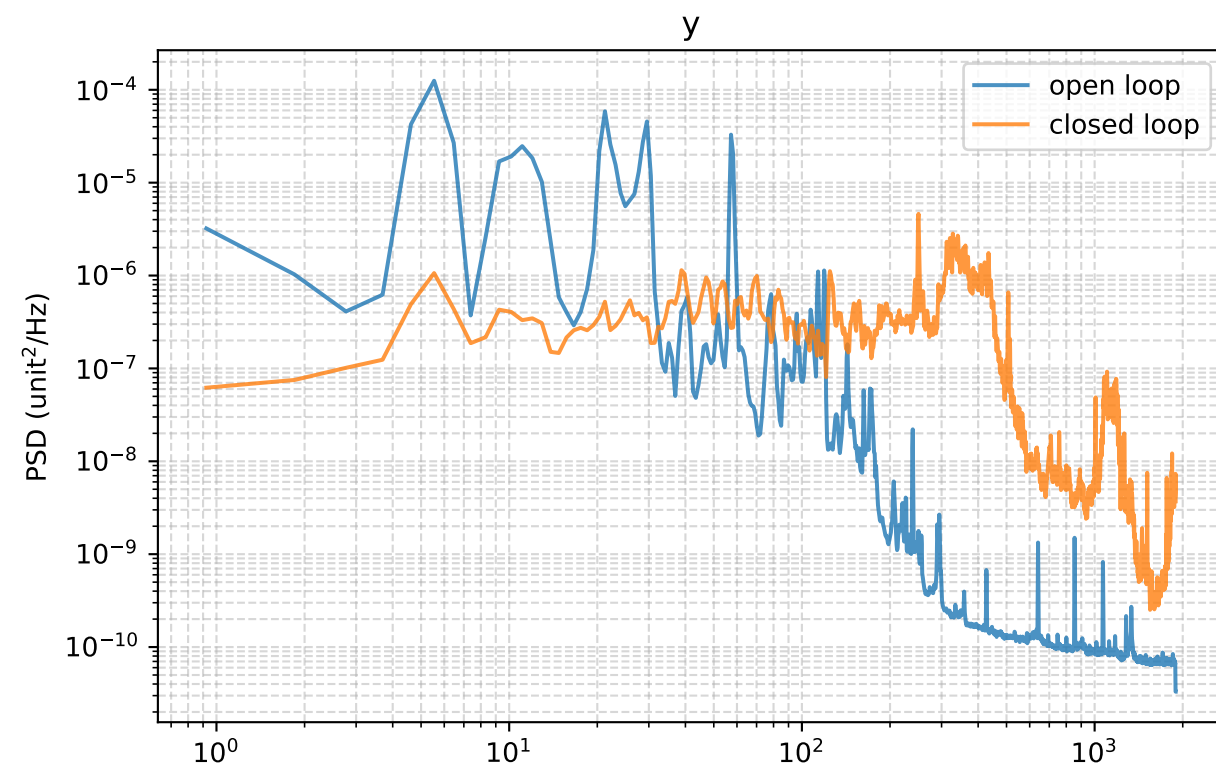


Closed-loop attenuation



Run configuration and per-band RMS

Started: 2026-07-17 12:34:10
pixel_scale: 15.5 px per output unit
y: mean -0.870 -> -0.001 px; RMS about mean 0.488 -> 0.285 px (open -> closed)
x: mean +1.888 -> +0.002 px; RMS about mean 0.689 -> 0.469 px (open -> closed)

Config used:
ATTENUATION11 FSP RETEST -- new laser alignment, camera rate, and current Bode plant gains
Same continuously adapting full-band FSP as the successful attenuation11 run: broad_order=512, broad_mu=.03, start_delay=2 s, ramp=10 s; no frozen-weights test mode or safety-trip tuning
Camera/CoM match steering_tuned.json: 32x32 ROI, exposure 50, gain 100, 20x20 tracked window, threshold 22; coarse offsets 1.95/2.02 V
Current Bode plots fsm_bodex_716/fsm_bodey_716 give low-frequency normalized gains Kx=.868 and Ky=.630
Loop delay from the bode phase fits (5-300 Hz slope + 1 hidden frame): x 1.484 ms = 5.62 fr, y 1.681 ms = 6.37 fr at 3788 Hz. PER-AXIS delays: y global 6+0.0 (validated), x override 5+0.62 (bode fit; 5x latency runs 21:22-21:26 give departure 1.06 ms / 50% 1.32 ms = 5.00 fr every run). Departure-latency 4.02 fr was 1.6-2.3 fr short and caused the 310 Hz spiral of 07-16 20:55
All remaining observer and modal parameters match attenuation11; output is reported at 15.5 px per normalized unit
refs: Kulcsar 2006 / Petit 2008 / Meimon 2010 / Correia 2010

RMS contributed by each 10 Hz band (px)

band (Hz)	y open	y closed	x open	x closed
0-10	0.2207	0.0266	0.1804	0.0584
10-20	0.1324	0.0257	0.2646	0.0508
20-30	0.2286	0.0305	0.3731	0.0532
30-40	0.0567	0.0355	0.0708	0.0389
40-50	0.0201	0.0370	0.0311	0.0627
50-60	0.1171	0.0347	0.1365	0.0452
60-70	0.0188	0.0365	0.0188	0.0807
70-80	0.0223	0.0317	0.0335	0.0405
80-90	0.0152	0.0334	0.1072	0.0528
90-100	0.0189	0.0259	0.0149	0.0663
100-110	0.0226	0.0243	0.0211	0.0428
110-120	0.0309	0.0196	0.0298	0.0396
120-130	0.0144	0.0376	0.0232	0.0436
130-140	0.0082	0.0280	0.0159	0.0426
140-150	0.0118	0.0250	0.0269	0.0446
150-160	0.0056	0.0308	0.0094	0.0555
160-170	0.0070	0.0235	0.0086	0.0457
170-180	0.0080	0.0210	0.0119	0.0432
180-190	0.0027	0.0291	0.0052	0.0414
190-200	0.0019	0.0319	0.0034	0.0342
200-210	0.0027	0.0275	0.0028	0.0271
210-220	0.0021	0.0288	0.0024	0.0282
220-230	0.0023	0.0281	0.0024	0.0278
230-240	0.0032	0.0259	0.0021	0.0276
240-250	0.0019	0.0492	0.0020	0.0466
250-260	0.0015	0.0535	0.0020	0.0578
260-270	0.0010	0.0294	0.0017	0.0664
270-280	0.0010	0.0255	0.0016	0.0764
280-290	0.0013	0.0260	0.0016	0.0674
290-300	0.0017	0.0366	0.0016	0.0680
total 0-300	0.3733	0.1724	0.5318	0.2805